

**Name:** White, Rice, & Spice Team**Date:** 3/9/26**Class:** Design for Engineers e4usa

# MyDesign Portfolio

## Element A: Presentation and Justification of the Problem

### Problem Statement and Details (A1, A2, A5)

Bathroom sinks often have standing water from handwashing. When soap dispensers sit directly on the counter, water collects underneath them. Moist environments allow bacteria and mold to grow quickly. The bottom of the bottle can become contaminated, and when people touch it to pump soap, germs can spread. In public bathrooms, this risk increases because more people use the same soap dispensers each day. The bacteria present will be gram-negative bacteria like *Serratia marcescens*, *Klebsiella pneumoniae*, and *Pseudomonas aeruginosa*. These can cause skin, eye, or urinary tract infections.

### Stakeholder Concerns and Connections (A3, A4)

1. Janitors are a primary stakeholder because they are responsible for supplying residents and businesses with soap dispensers, and they are responsible for cleaning up. As a result, they are always looking for ways to efficiently clean up messes. We can contact the janitors through their business representatives.
2. Business owners are a primary stakeholder because this could be a profitable investment as it could benefit their employees, and as the invention grows, so does their profit. We can contact these venture capitalists through their websites and message them on LinkedIn.
3. School Students are also stakeholders. When one person gets sick in a school, everyone else is more likely chance of getting sick due to being so crammed in school. By making soap dispensers safer and carrying less germs, school students will be healthier.

Students are secondary stakeholders, as these inventions will affect the cleanliness and overall hygiene of their bathrooms. They are not investors themselves, but the invention will have a direct impact on them. Employees are in the same situation as students, as they directly benefit from this but are not the ones investing in or cleaning them.

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### Problem Importance (A6)

Soap dispensers can ironically hold unsafe levels of bacteria that can lead to negative health outcomes. For example, a U.S. Library of Medicine article found that unmaintained soap dispensers frequently have unsafe levels of bacteria. They are unsafe because they can enter the human body, particularly in older people, and cause infections that can lead to amputation or death. In regular people, the germs and bacteria can cause high fevers and make people sick by irritating the lungs. Finally, there is a historical precedent for our problem: OSHA has rules in place to prevent hand washing puddles from the floor, as it causes safety risks for customers and employees, as bacterial growth skyrockets when humidity is mixed with water

#### References:

"Falls and Fractures in Older Adults: Causes and Prevention | National Institute on Aging." Edited by Editors of [nia.nih.gov](http://nia.nih.gov), *National Institute on Aging*, 12 Sept. 2022, [www.nia.nih.gov/health/falls-and-falls-prevention/falls-and-fractures-older-adults-causes-and-prevention](http://www.nia.nih.gov/health/falls-and-falls-prevention/falls-and-fractures-older-adults-causes-and-prevention).

Lucassen, Ralf, et al. "A Loophole in Soap Dispensers Mediates Contamination with Gram-Negative Bacteria." *MicrobiologyOpen*, U.S. National Library of Medicine, 29 Sept. 2023, [pmc.ncbi.nlm.nih.gov/articles/PMC10541457/](https://pmc.ncbi.nlm.nih.gov/articles/PMC10541457/).

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